Visitor Mobility and Behavior Analysis in Logan Pass

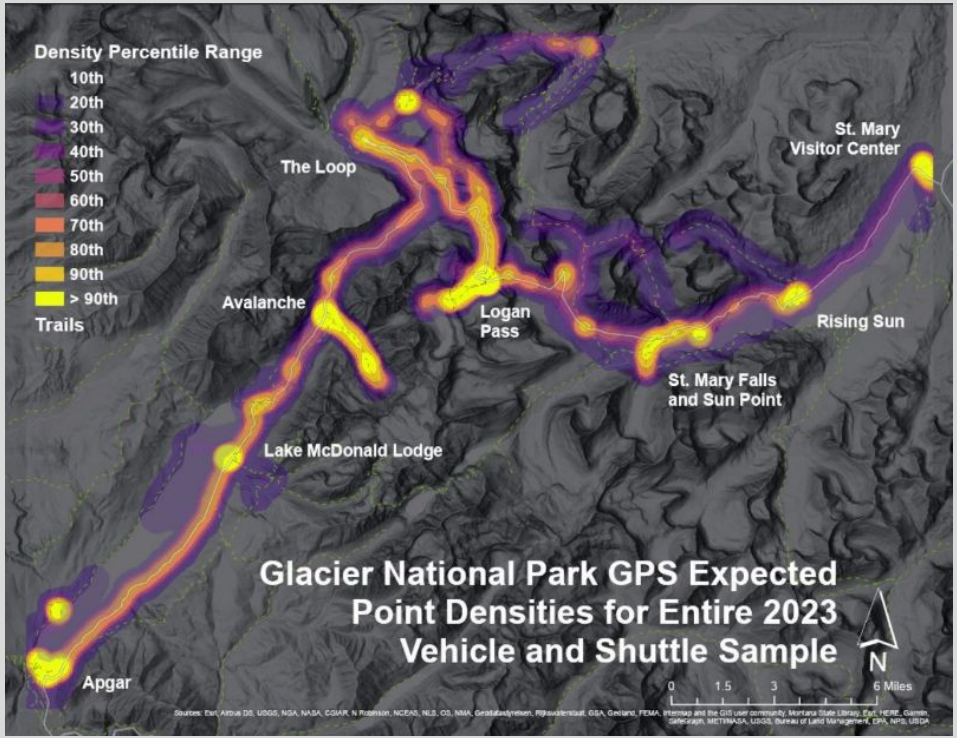
Leo Borsari, Ethan Henry, Connor Howard,
Cam Wilding, Daniel Zhang

10/10/2024

WPI

NATIONAL PARK SERVICE

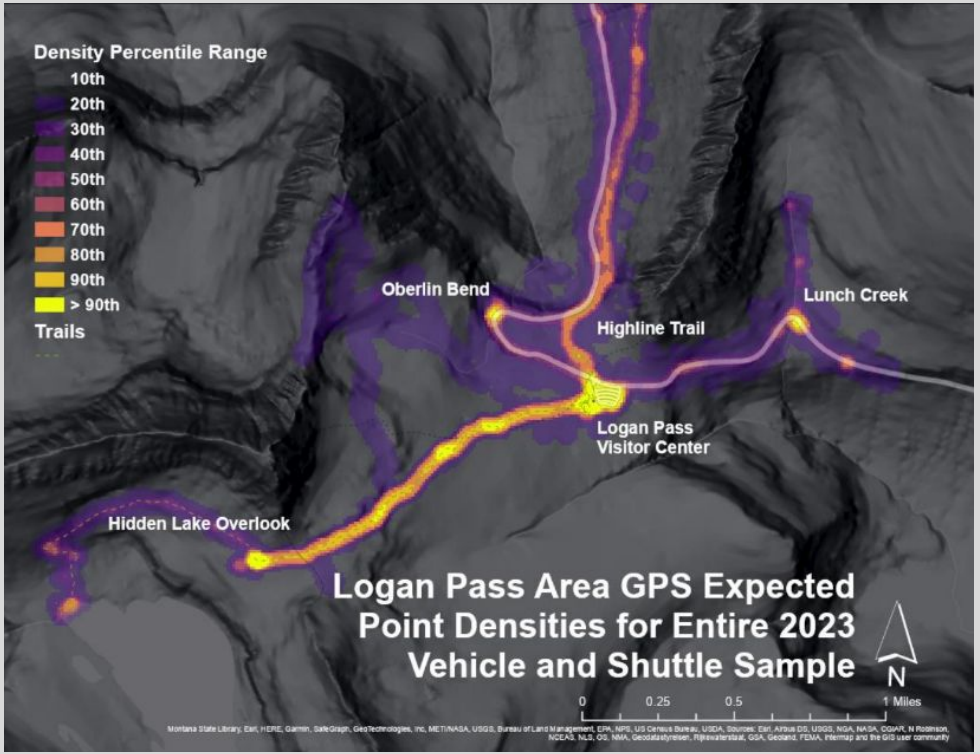
As visitation numbers swell, Glacier National Park struggles to understand visitor behavior and itineraries.



Heat map of anticipated visitor density locations, generated by University of Montana study. (Nayak et al., 2024, pg. 31).

University of Montana GNP Visitor Flow Study

- Heatmaps for visitor density in Logan Pass.
- Visitor travel to and from Logan Pass (shuttles vs. private vehicles).
- Time visitors spend at Logan Pass.



Heat map of anticipated visitor density at Logan Pass, generated by University of Montana study. (Nayak et al., 2024, pg. 38).

Limitations of Past Research

Lacks visitor behavior data, visitor itineraries specific to Logan Pass, and locational context.

Developed Area	Mean Stay Time	Median Stay Time	Count
Apgar	0:58	0:34	69
Lake McDonald lodge	0:31	0:09	331
Avalanche	0:25	0:11	316
Logan Pass	0:26	0:17	305
Rising Sun	0:43	0:07	236
St. Mary	3:31*	0:34	134
Note: Most visitors stay less than 1 hour in this developed area. Some visitors who stayed more than 8 hours are likely to be visitors staying overnight at the St. Mary campground.			

Chart showing the amount of time visitors spend at developed locations (Nayak et al., 2024, pg. 13).

Stops	Count	%
Logan Pass	50	56%
Avalanche	29	32%
Lake McDonald	6	7%
St. Mary Falls	3	3%
Sunrift gorge	1	1%
Rising Sun	1	1%
Sun Point	1	1%
Grant Total	90	100%

Chart ordering west shuttle stops by popularity (Nayak et al., 2024, pg. 24).

Supplied N = 454

Cleaned N = 350

GOAL:

To assist Glacier National Park better manage crowding in Logan Pass.

1

Find trends in visitor itineraries in Logan

Use ArcGIS to process analyze GPS data.

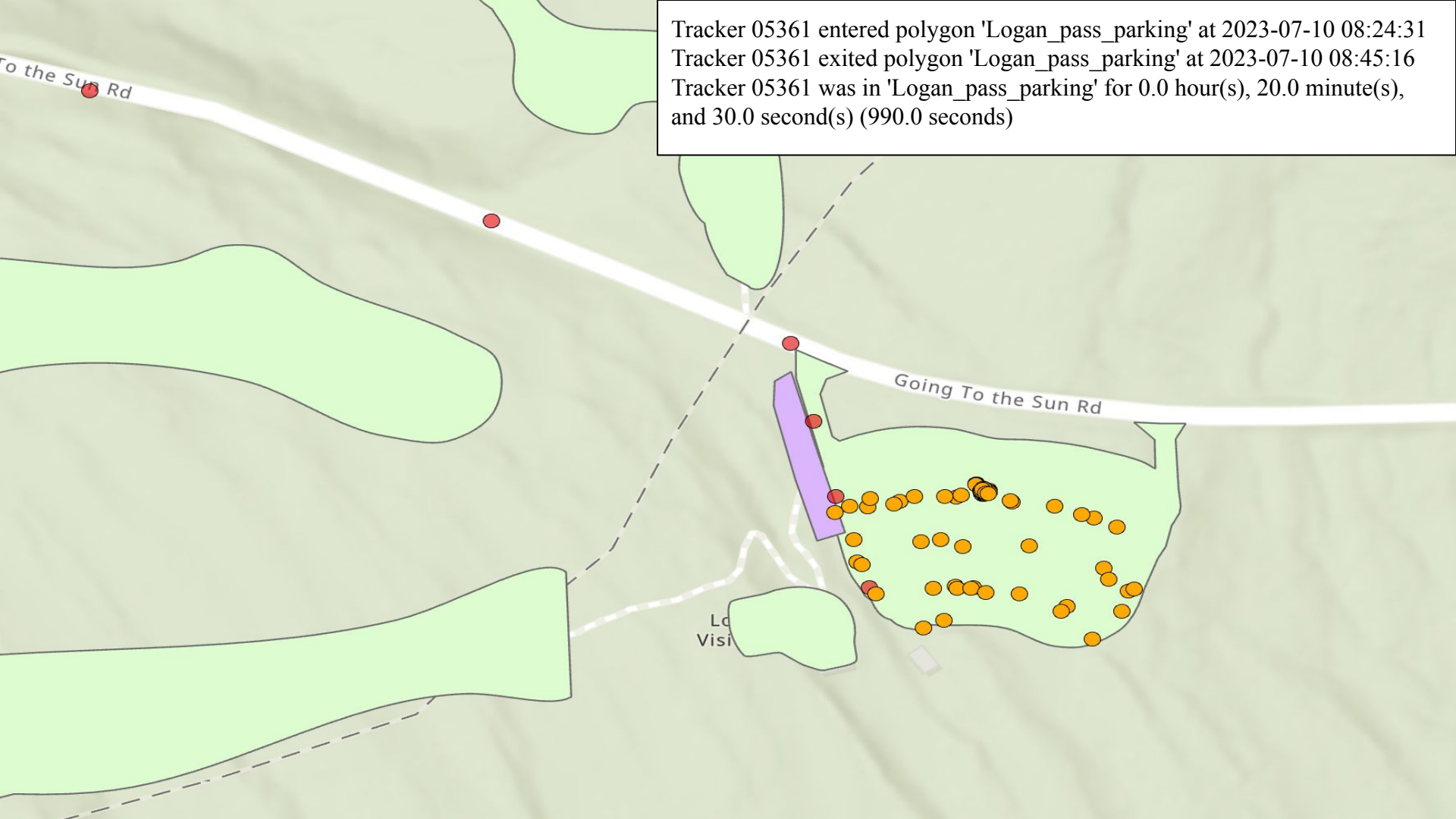
Identify typical visitor itineraries.

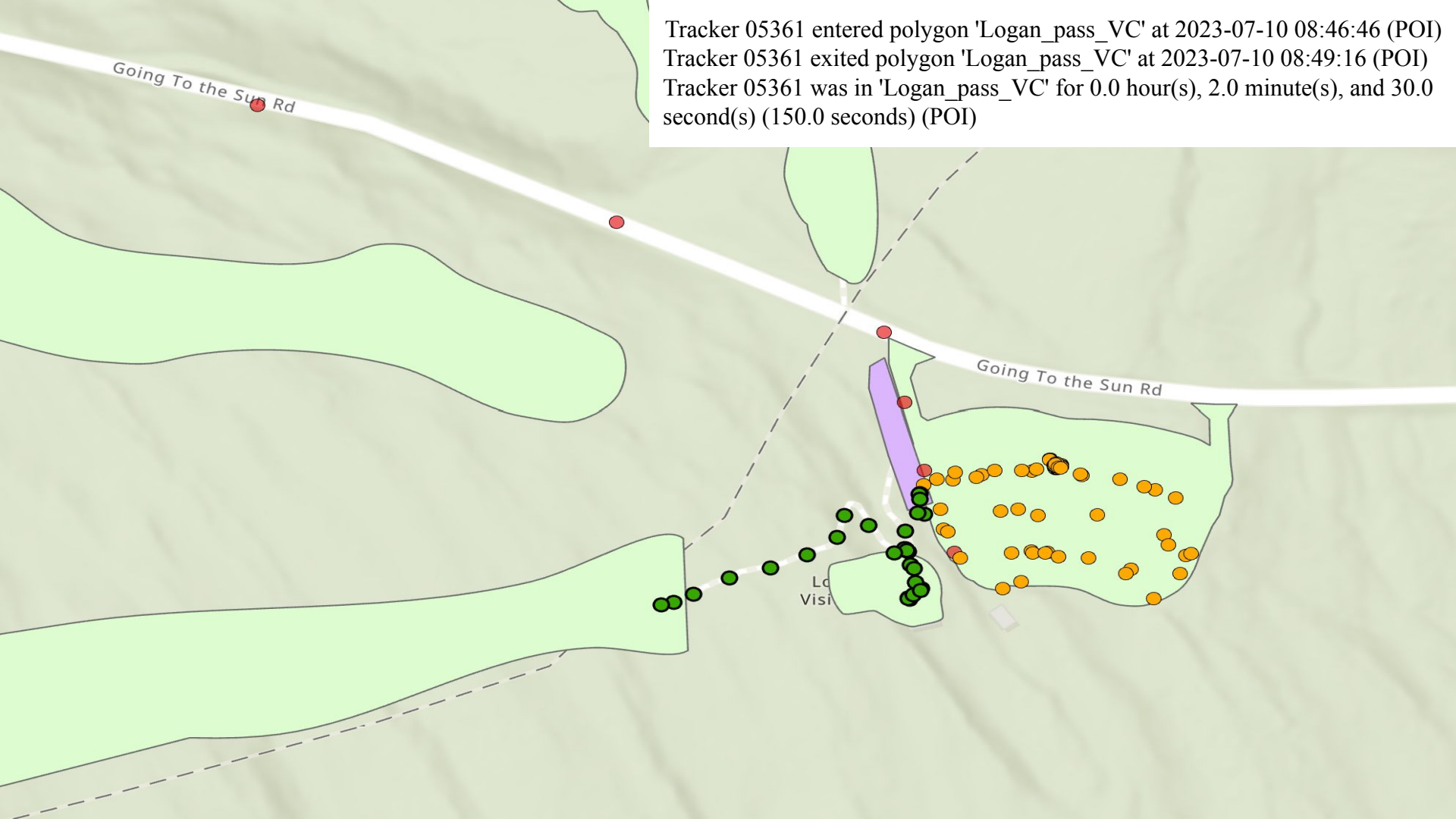
Develop visuals to show results.

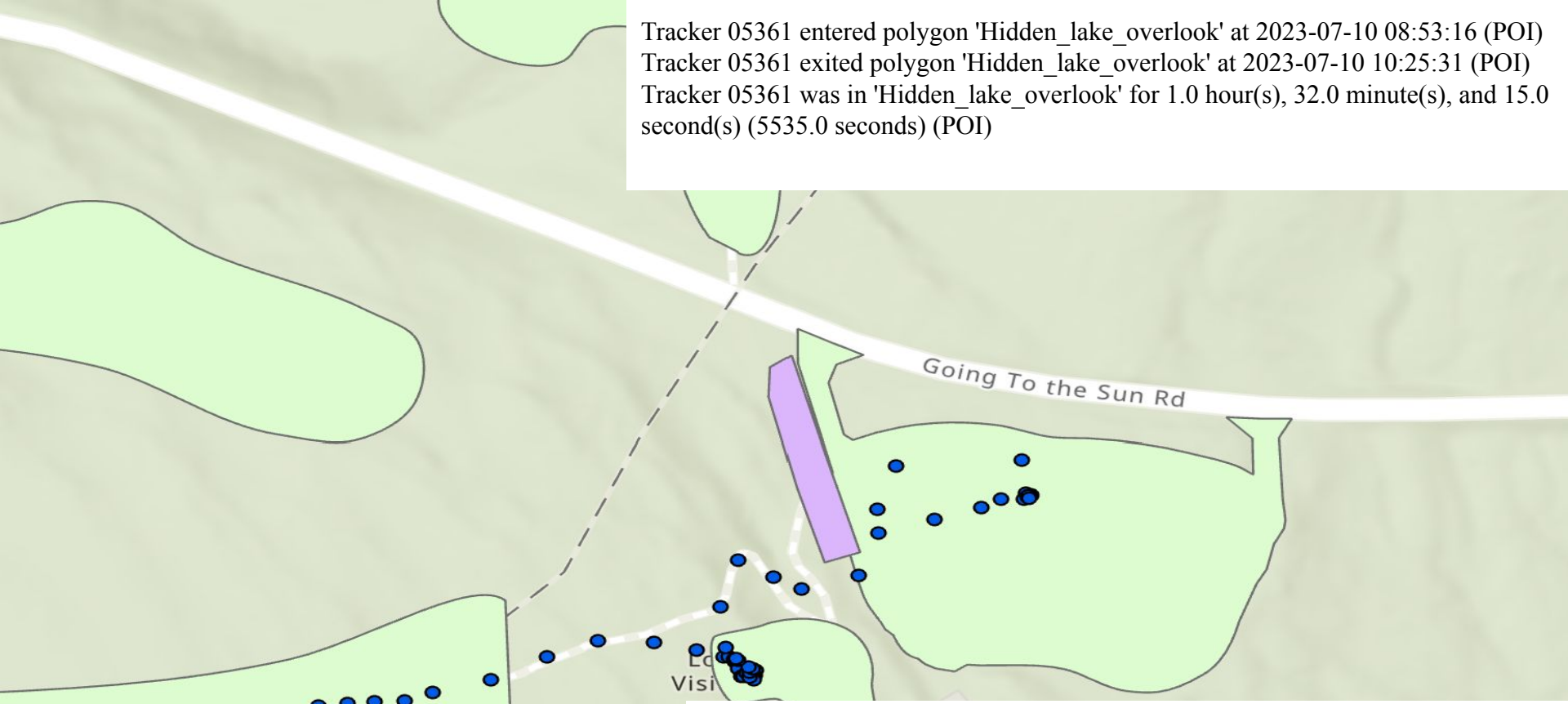
How did we display visitor trips throughout the park?

Tracker 05361 entered polygon 'Logan_pass_parking' at 2023-07-10 08:24:31 (POI)
Tracker 05361 exited polygon 'Logan_pass_parking' at 2023-07-10 08:28:31 (POI)
Tracker 05361 was in 'Logan_pass_parking' for 0.0 hour(s), 4.0 minute(s), and 0.0 second(s) (240.0 seconds) (POI)
Tracker 05361 entered polygon 'Logan_pass_parking' at 2023-07-10 08:28:46 (POI)
Tracker 05361 exited polygon 'Logan_pass_parking' at 2023-07-10 08:45:16 (POI)
Tracker 05361 was in 'Logan_pass_parking' for 0.0 hour(s), 16.0 minute(s), and 30.0 second(s) (990.0 seconds) (POI)
Tracker 05361 entered polygon 'Logan_pass_VC' at 2023-07-10 08:46:46 (POI)
Tracker 05361 exited polygon 'Logan_pass_VC' at 2023-07-10 08:49:16 (POI)
Tracker 05361 was in 'Logan_pass_VC' for 0.0 hour(s), 2.0 minute(s), and 30.0 second(s) (150.0 seconds) (POI)
Tracker 05361 entered polygon 'Logan_pass_VC' at 2023-07-10 08:51:16 (POI)
Tracker 05361 exited polygon 'Logan_pass_VC' at 2023-07-10 08:51:31 (POI)
Tracker 05361 was in 'Logan_pass_VC' for 0.0 hour(s), 0.0 minute(s), and 15.0 second(s) (15.0 seconds) (POI) (CLIP)
Tracker 05361 entered polygon 'Hidden_lake_overlook' at 2023-07-10 08:53:16 (POI)
Tracker 05361 exited polygon 'Hidden_lake_overlook' at 2023-07-10 10:25:31 (POI)
Tracker 05361 was in 'Hidden_lake_overlook' for 1.0 hour(s), 32.0 minute(s), and 15.0 second(s) (5535.0 seconds) (POI)
Tracker 05361 entered polygon 'Logan_pass_VC' at 2023-07-10 10:26:31 (POI)
Tracker 05361 exited polygon 'Logan_pass_VC' at 2023-07-10 10:35:16 (POI)
Tracker 05361 was in 'Logan_pass_VC' for 0.0 hour(s), 8.0 minute(s), and 45.0 second(s) (525.0 seconds) (POI)
Tracker 05361 entered polygon 'Logan_pass_parking' at 2023-07-10 10:36:16 (POI)
Tracker 05361 exited polygon 'Logan_pass_parking' at 2023-07-10 10:40:46 (POI)
Tracker 05361 was in 'Logan_pass_parking' for 0.0 hour(s), 4.0 minute(s), and 30.0 second(s) (270.0 seconds) (POI)


Highline Trail
Tracker 05361 entered polygon Logan Pass Parking at 2023-07-10 08:24:31
Going To the Sun Rd
Logan Pass Parking
Hidden Lake Overlook
Logan Pass Visitor Center

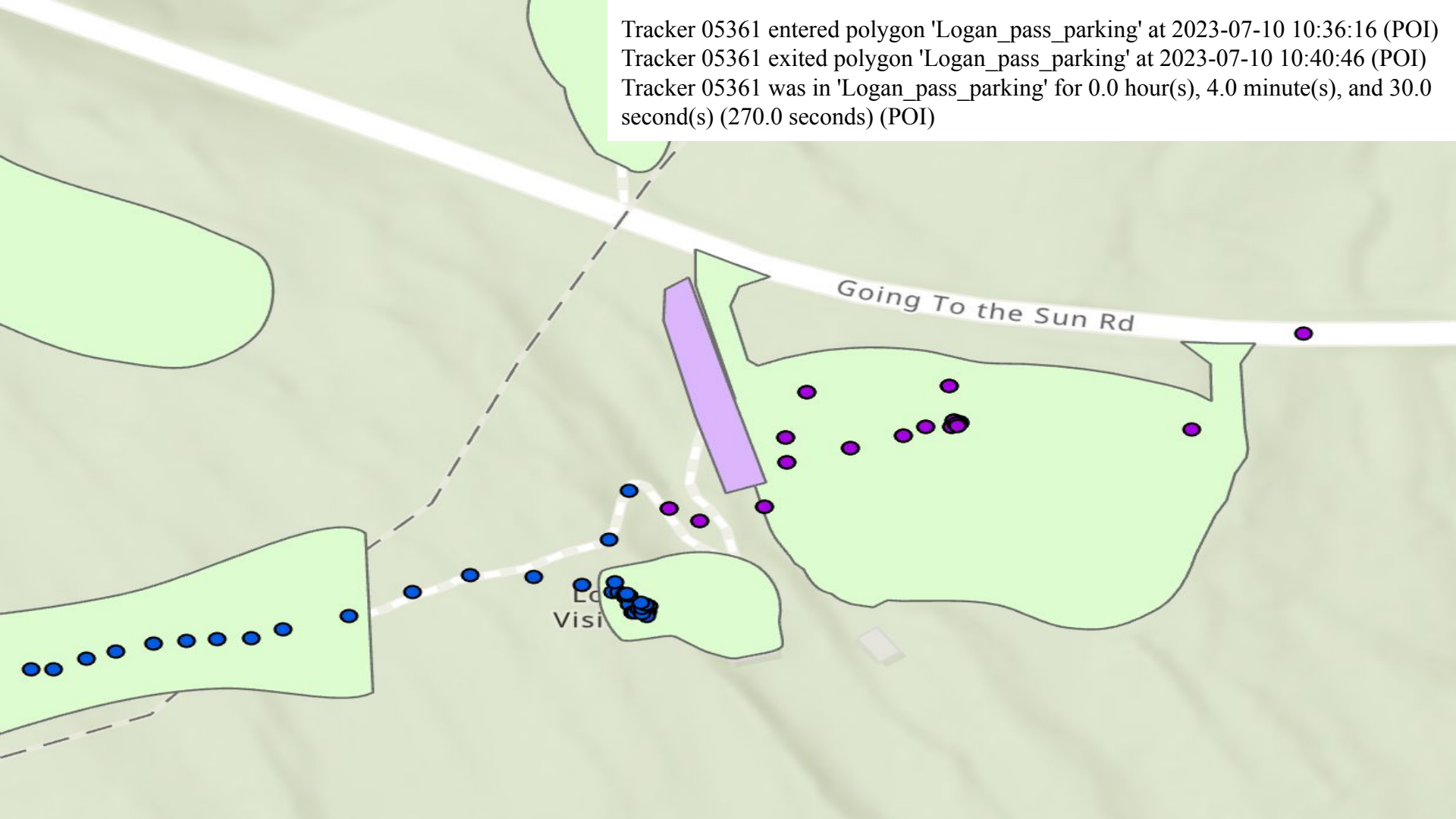

Tracker 05361 entered polygon 'Logan_pass_parking' at 2023-07-10 08:24:31
Tracker 05361 exited polygon 'Logan_pass_parking' at 2023-07-10 08:45:16
Tracker 05361 was in 'Logan_pass_parking' for 0.0 hour(s), 20.0 minute(s), and 30.0 second(s) (990.0 seconds)
Going To the Sun Rd


Tracker 05361 entered polygon 'Logan_pass_VC' at 2023-07-10 08:46:46 (POI)
Tracker 05361 exited polygon 'Logan_pass_VC' at 2023-07-10 08:49:16 (POI)
Tracker 05361 was in 'Logan_pass_VC' for 0.0 hour(s), 2.0 minute(s), and 30.0 second(s) (150.0 seconds) (POI)
Going To the Sun Rd
Going To the Sun Rd



Tracker 05361 entered polygon 'Hidden_lake_overlook' at 2023-07-10 08:53:16 (POI)
Tracker 05361 exited polygon 'Hidden_lake_overlook' at 2023-07-10 10:25:31 (POI)
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Tracker 05361 exited polygon 'Logan_pass_parking' at 2023-07-10 10:40:46 (POI)
Tracker 05361 was in 'Logan_pass_parking' for 0.0 hour(s), 4.0 minute(s), and 30.0 second(s) (270.0 seconds) (POI)
Going To the Sun Rd

Hiking a Trail

Going to a Visitor Center
Going on a Shuttle

Looking for Parking

id	location_1
05361	Loop_parking
01031	Lake_macdon
01061	Lake_macdon
01241	Lake_macdon
01341	Lake_macdon
01371	Lake_macdon
01401	Lake_macdon
01501	Lake_macdon
01511	Lake_macdon
02011	Logan_pass_
02021	Logan_pass_
02031	Lake_macdon
02061	Trail_of_Ceda
02081	Apgar_VC_pa
02091	Apgar_VC_pa
02101	Apgar_VC_pa
02111	Apgar_VC_pa
02141	Logan_pass_



2

Evaluate park staff perceptions of the visitor experience.

Interview park staff to identify what attracts visitors to Logan Pass.

Qualify park staff understanding of common visitor itineraries.

3

Identify contextual conditions that shape travel itineraries

Conduct first-hand observations to define locational context at Logan Pass.

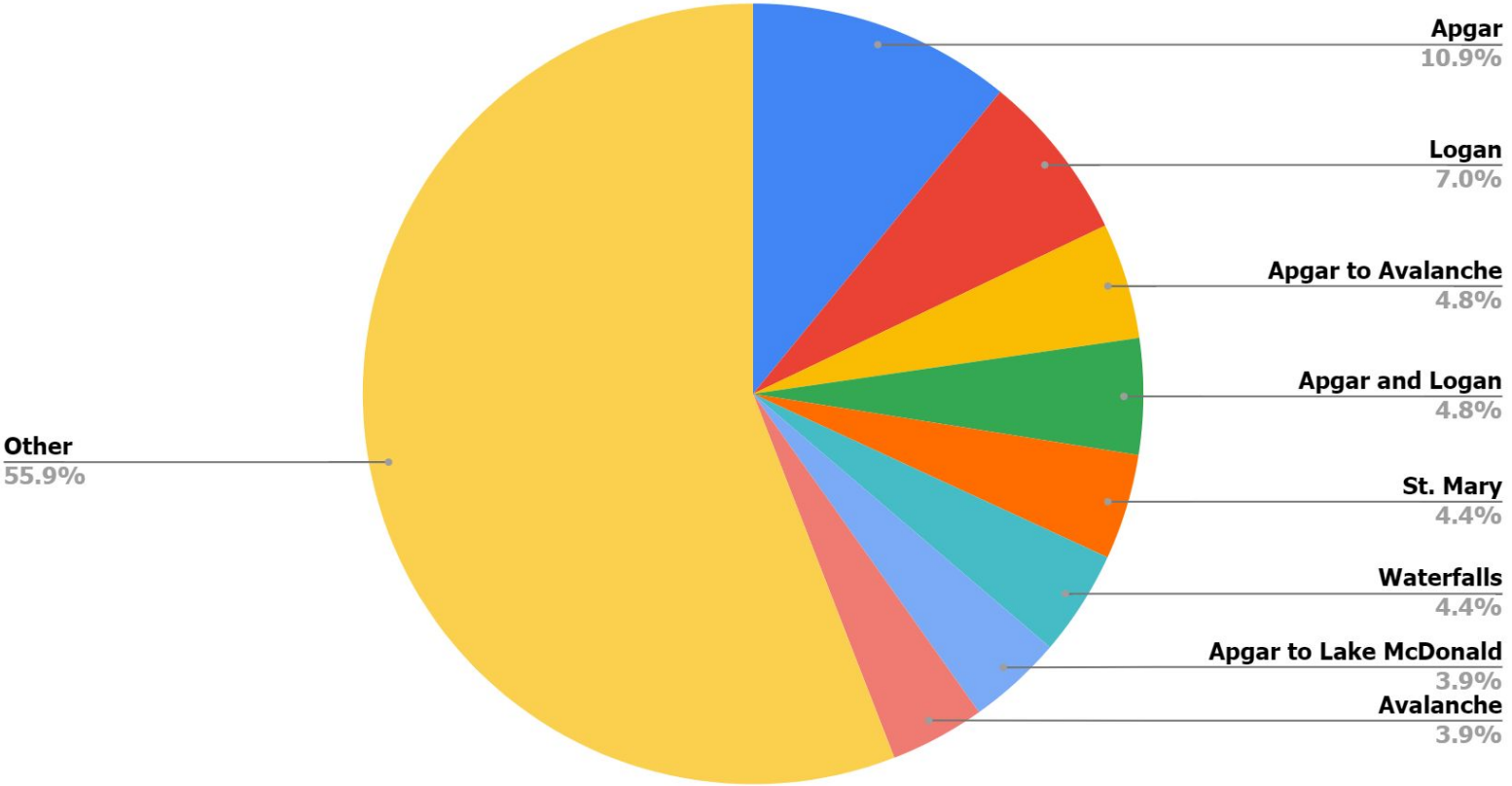
Verify geo-fenced areas and GPS data.

Results

Even the “most common” itineraries are comparatively small pieces of a larger whole.

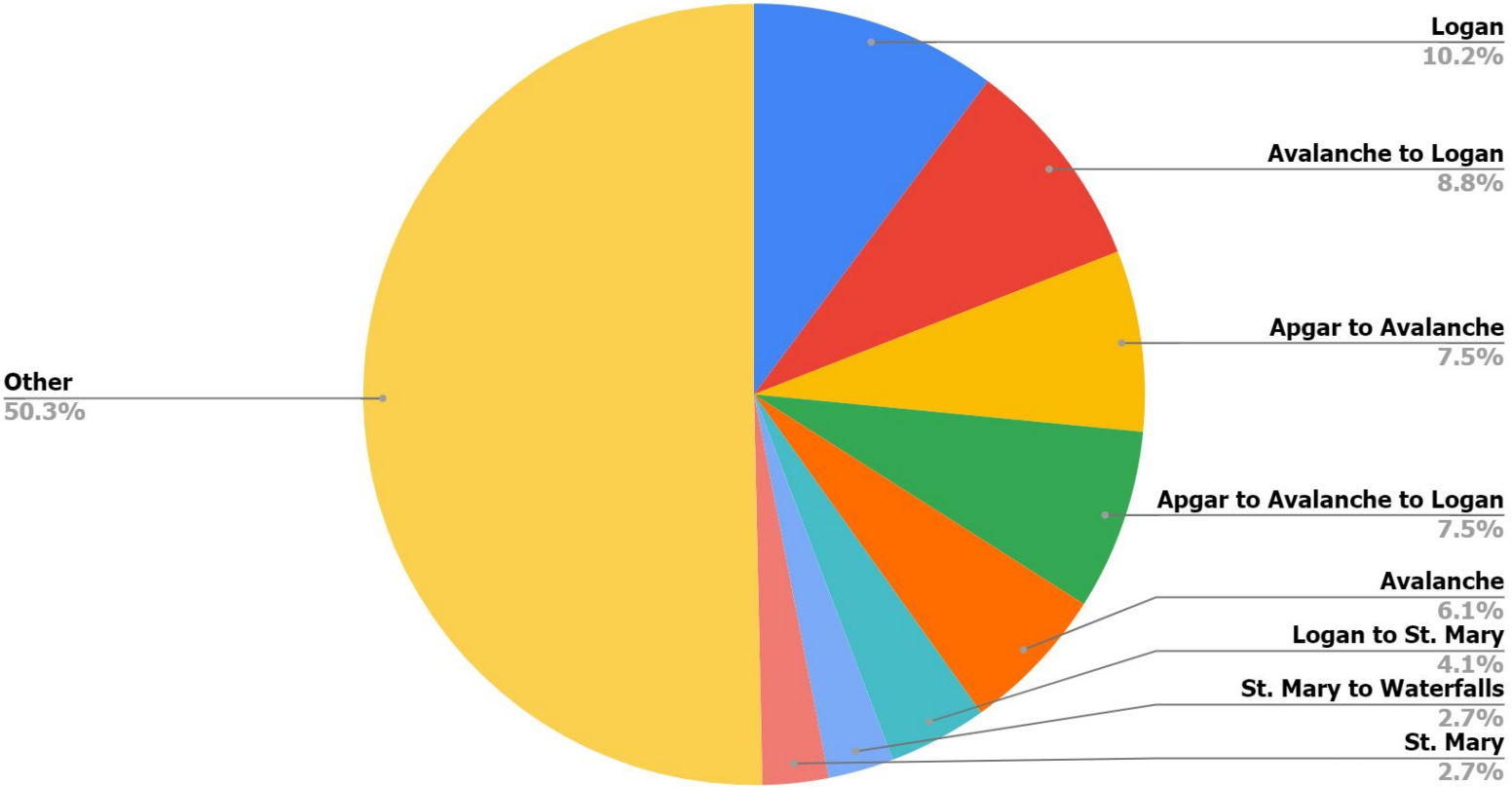
Visitor Itineraries Throughout the Park - Private Vehicle Users

(N = 229)



Visitor Itineraries Throughout the Park - Shuttle Users

(N = 119)



People go out of their way to visit Apgar.

Locations Visited During Trip	Private Vehicles (n=229)	Count for Private Vehicle (East) (n=94)	Count for Private Vehicle(West) (n=135)	After Managed Access (n=134)	Managed Access (n=95)
Apgar	**25**	**10**	**15**	**17**	**8**
Logan	**16**	**11**	**5**	**11**	**5**
Apgar Avalanche	**11**	**1**	**10**	**1**	**10**
Apgar Logan	**11**	**5**	**6**	**7**	**4**
StMary	**10**	**5**	**5**	**4**	**6**
Waterfalls	**10**	**10**	**0**	**8**	**2**
Apgar Lake McDonald	**9**	**1**	**8**	**2**	**7**
Avalanche	**9**	**2**	**7**	**4**	**5**
Other	**128**	**40**	**79**	**80**	**48**

Locations Visited During Trip	Private Vehicles (n=229)	Count for Private Vehicle (East) (n=94)	Count for Private Vehicle(West) (n=135)	After Managed Access (n=134)	Managed Access (n=95)
Apgar	**25**	**10**	**15**	**17**	**8**
Logan	**16**	**11**	**5**	**11**	**5**
Apgar Avalanche	**11**	**1**	**10**	**1**	**10**
Apgar Logan	**11**	**5**	**6**	**7**	**4**
StMary	**10**	**5**	**5**	**4**	**6**
Waterfalls	**10**	**10**	**0**	**8**	**2**
Apgar Lake McDonald	**9**	**1**	**8**	**2**	**7**
Avalanche	**9**	**2**	**7**	**4**	**5**
Other	**128**	**40**	**79**	**80**	**48**

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Apgar	**25**	**10**	**15**	**17**	**8**
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Apgar Logan	**11**	**5**	**6**	**7**	**4**
StMary	**10**	**5**	**5**	**4**	**6**
Waterfalls	**10**	**10**	**0**	**8**	**2**
Apgar Lake McDonald	**9**	**1**	**8**	**2**	**7**
Avalanche	**9**	**2**	**7**	**4**	**5**
Other	**128**	**40**	**79**	**80**	**48**

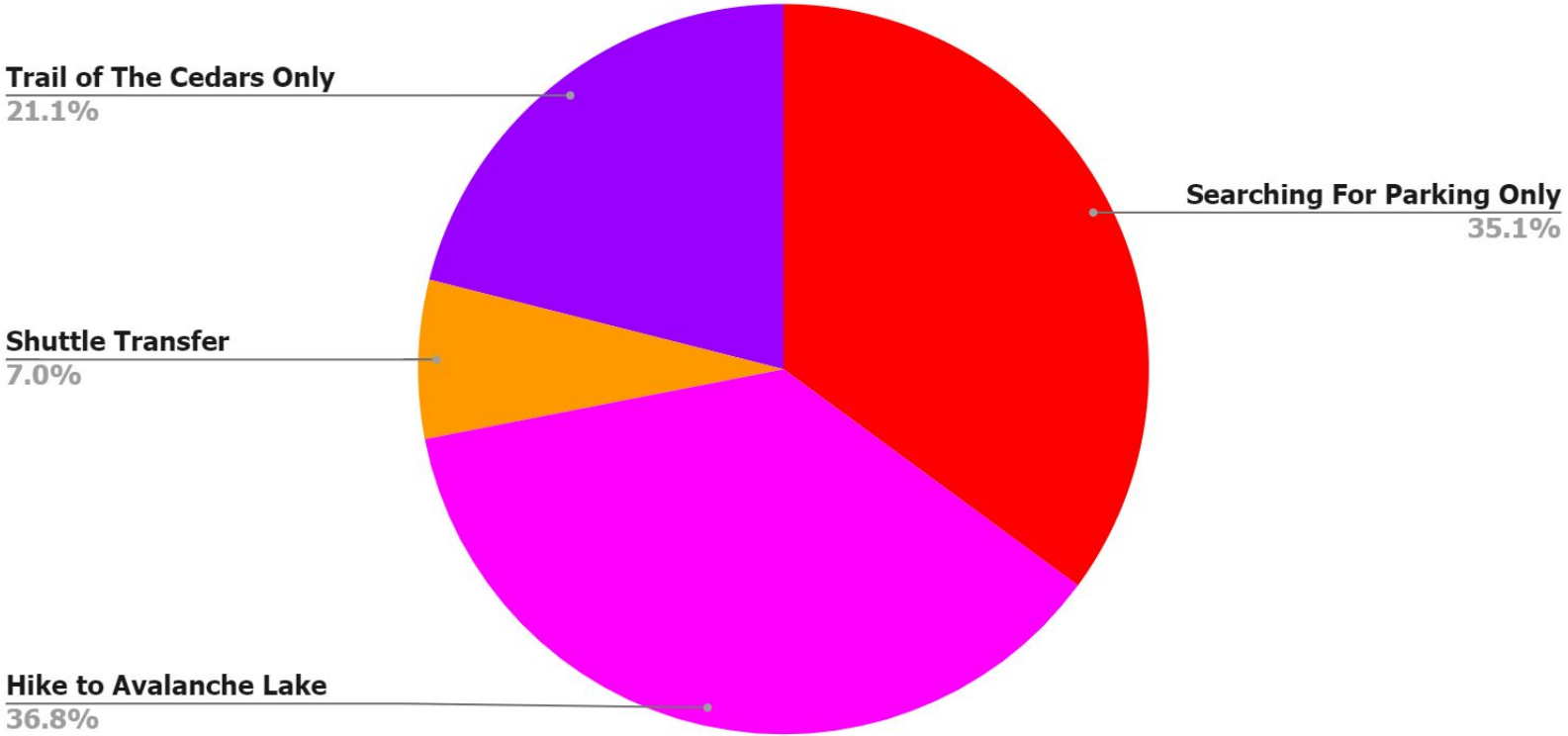
Locations Visited During Trip	Private Vehicles (n=229)	Count for Private Vehicle (East) (n=94)	Count for Private Vehicle(West) (n=135)	After Managed Access (n=134)	Managed Access (n=95)
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Apgar Logan	**11**	**5**	**6**	**7**	**4**
StMary	**10**	**5**	**5**	**4**	**6**
Waterfalls	**10**	**10**	**0**	**8**	**2**
Apgar Lake McDonald	**9**	**1**	**8**	**2**	**7**
Avalanche	**9**	**2**	**7**	**4**	**5**
Other	**128**	**40**	**79**	**80**	**48**

Visitors are parking at Avalanche to take the GTSR shuttle to other locations.

Behavior - Logan	Color	Notes
Hidden Lake Overlook (HLO)		The Hidden Lake Overlook is only one half of the Hidden Lake trail. Only 1 out of 78 total hikers completed the whole thing.
Highline Trail (HT)		The Highline is one of Glacier's longest hikes on the GSTR.
Both Hidden Lake Overlook and Highline Trail		Some adventurous hikers combined both hikes.
Non - Hiker		These are the people who traveled to Logan Pass and the visitor center but did not enter any trails.
Shuttle Transfer		Logan Pass is the East/West shuttle transfer point. Some visitors use Logan to go to shuttle stops at the other side of the park.
Searching for		This data is from 7am onward in July so Logan pass is already full

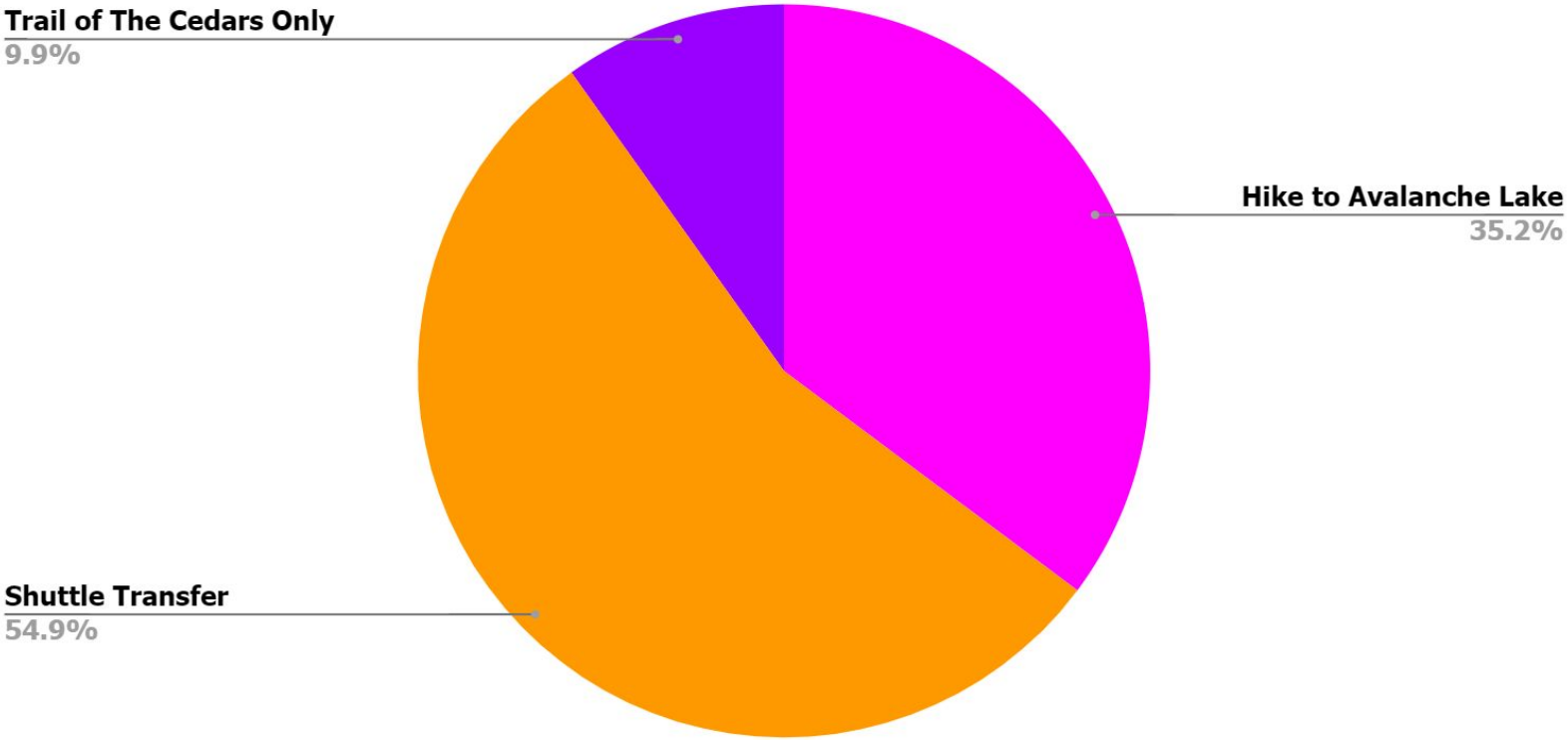
Visitor Behavior at Avalanche for ALL Private Vehicle Arrivals

(N = 57)

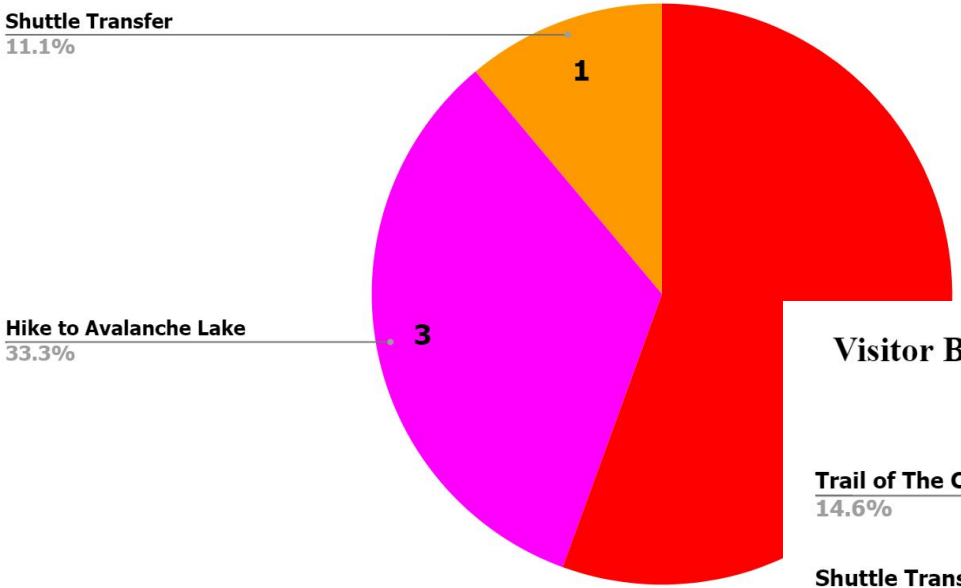


Visitor Behavior at Avalanche for Shuttle Arrivals

(N = 71)

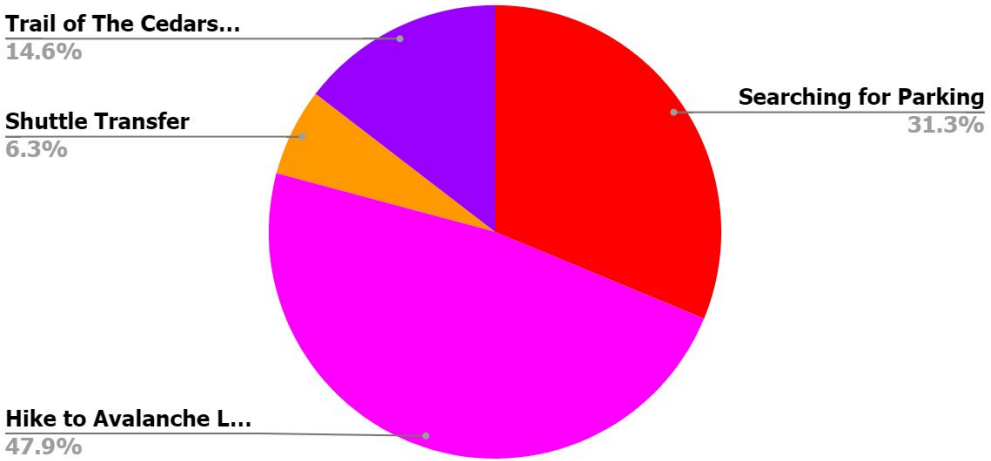


Visitor Behavior at Avalanche After Managed Access (Private Vehicle Arrivals)



Visitor Behavior at Avalanche During Managed Access (Private Vehicle Arrivals)

(N = 48)



Though most people visit more than one location within the park, this behavior is more commonly found in hikers than in non-hikers.

For private vehicle users *:*

76.7% of hikers and 58.3% non-hikers visited more than one location.

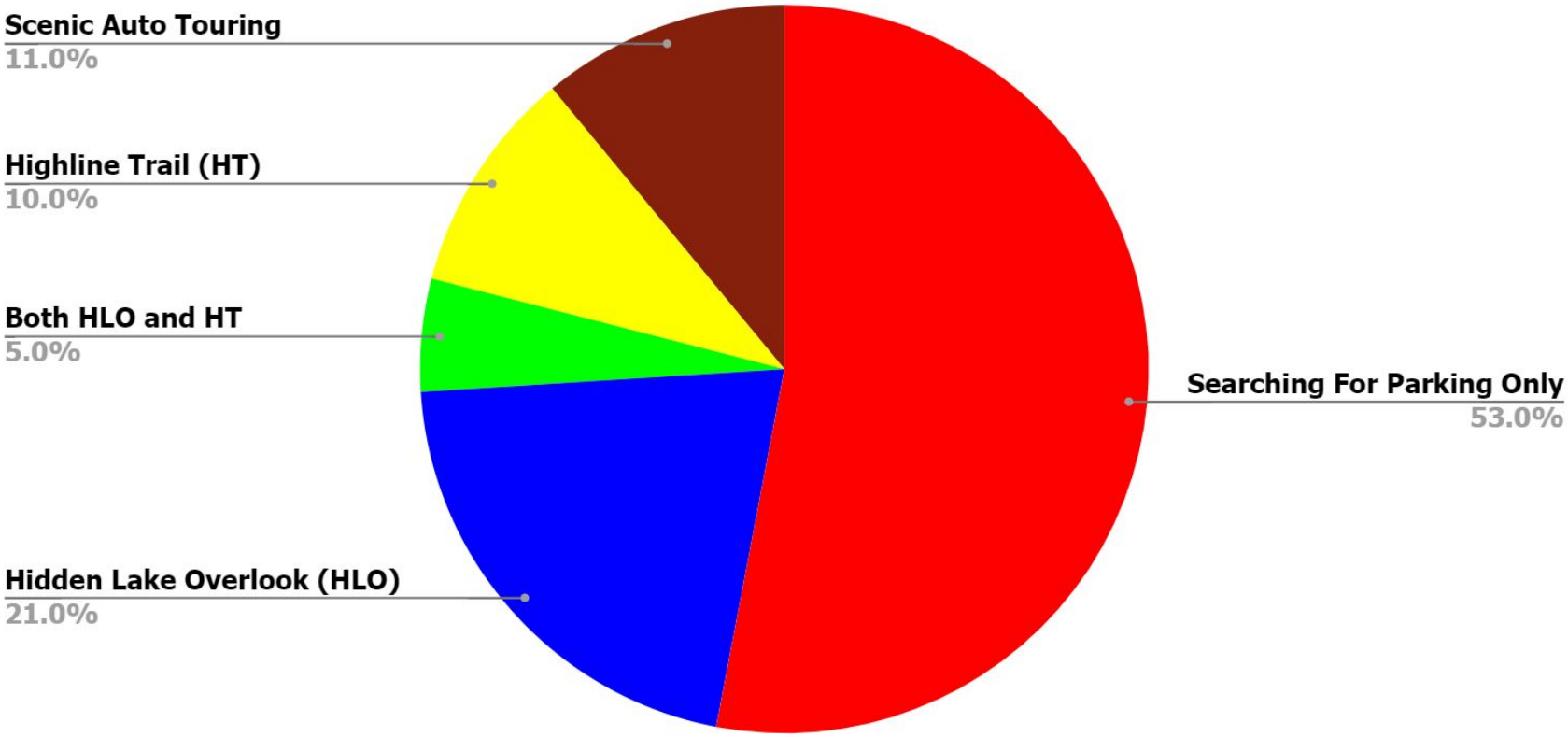
For shuttle users *:*

93.3% of shuttle hikers and 46.7% of shuttle non-hikers visited more than one location.

53% of tracked private vehicle users end their trip **without finding parking in Logan Pass.**

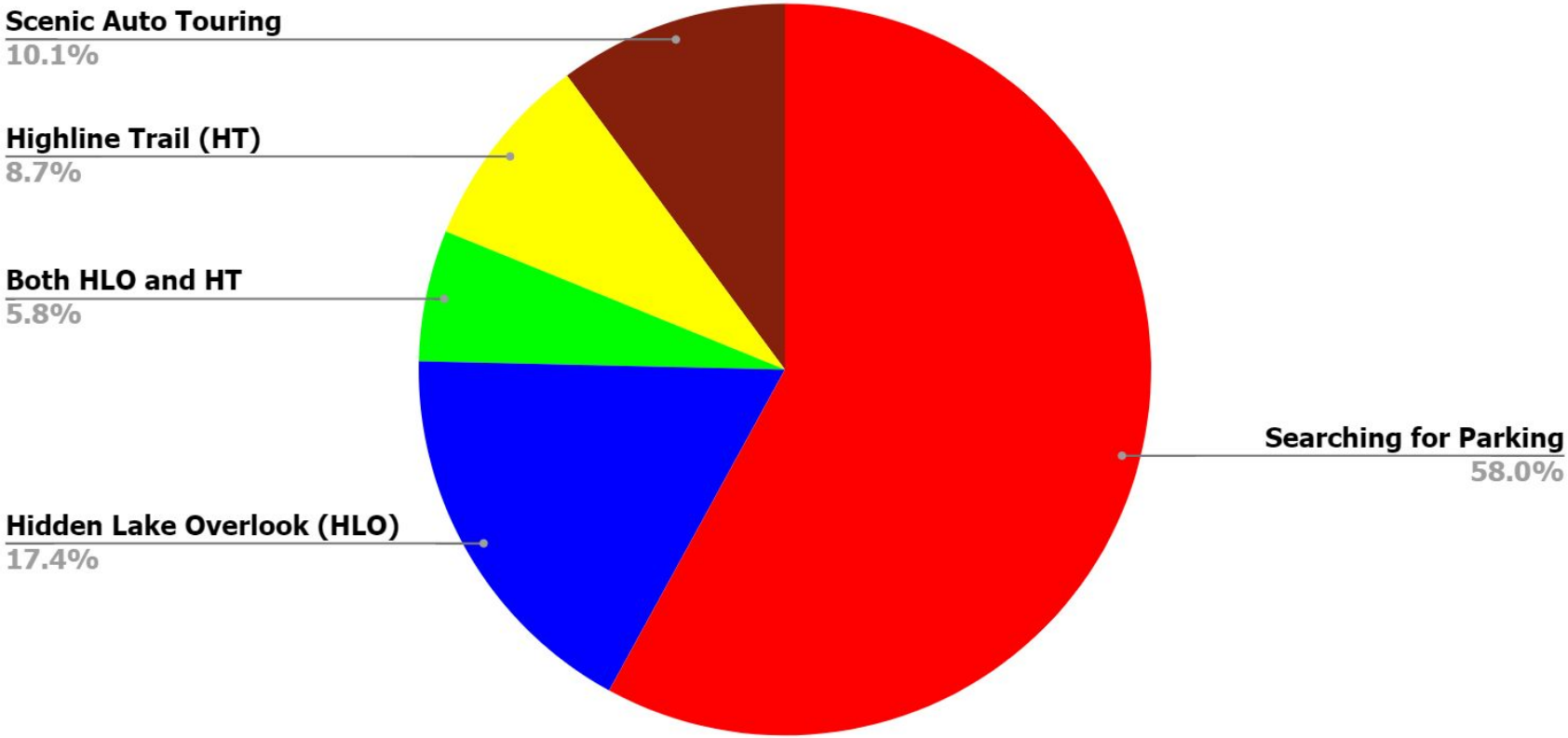
Visitor Behavior at Logan Pass for ALL Private Vehicles Arrivals

(N = 100)



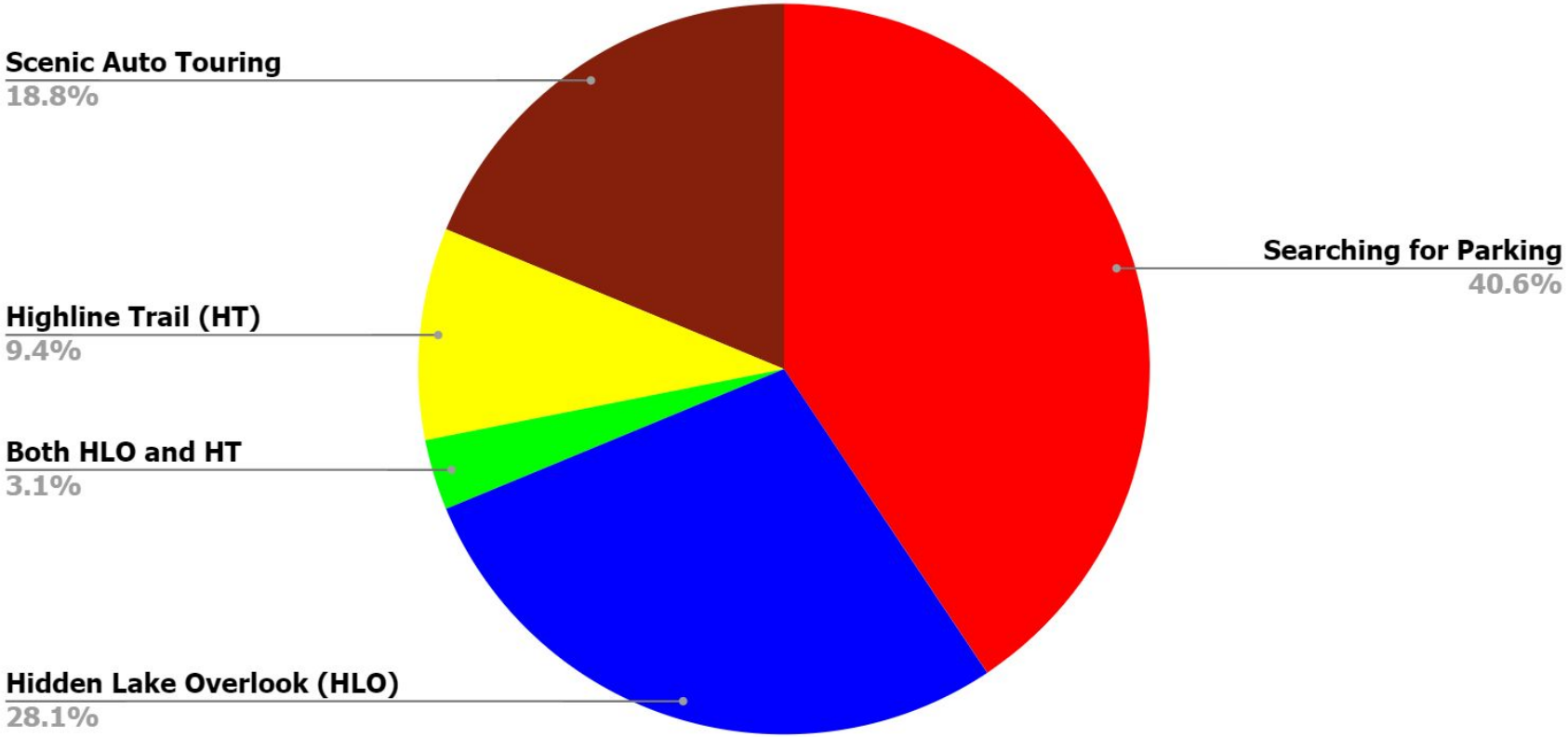
Visitor Behavior at Logan Pass During Managed Access (Private Vehicle Arrivals)

(N = 69)



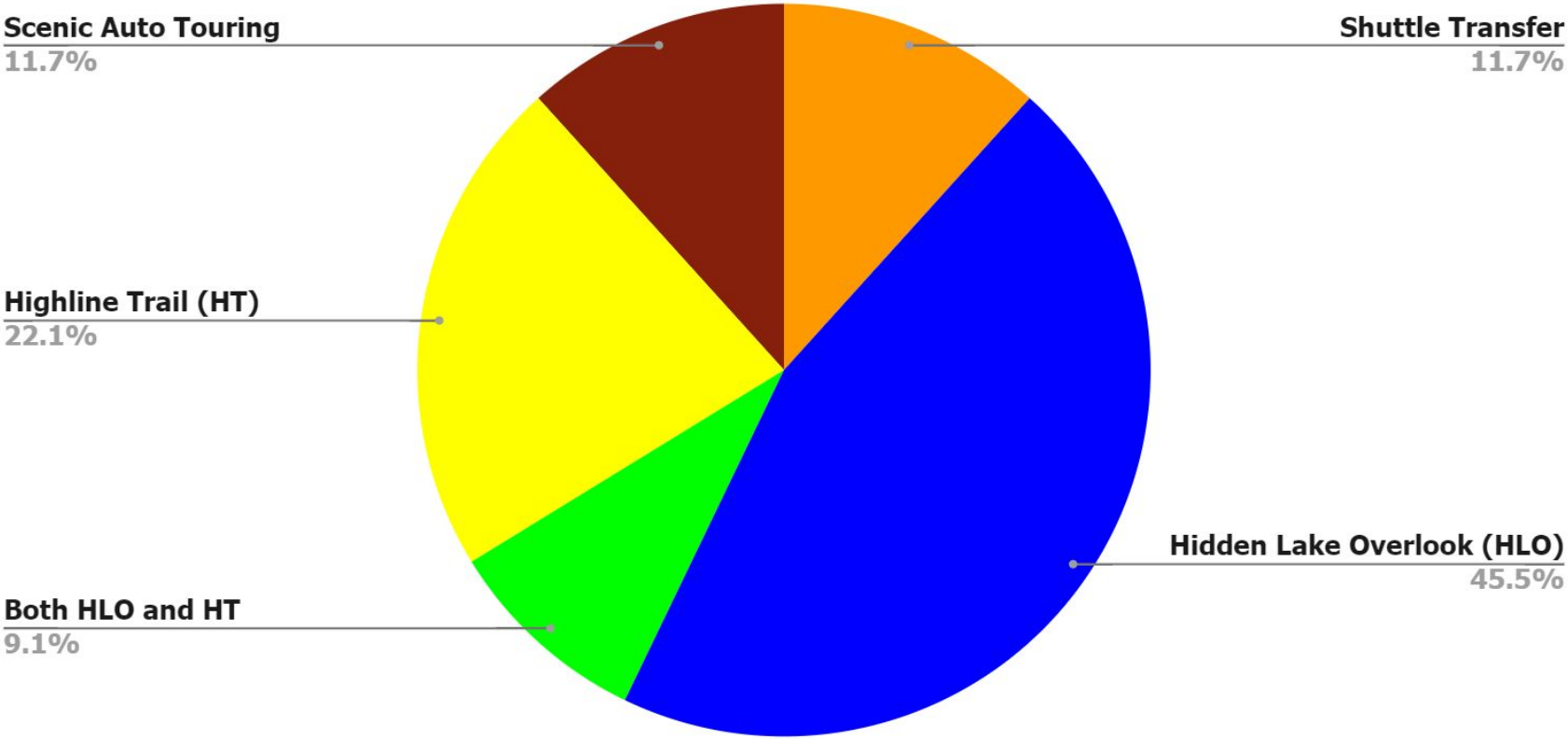
Visitor Behavior at Logan Pass After Managed Access (Private Vehicle Arrivals)

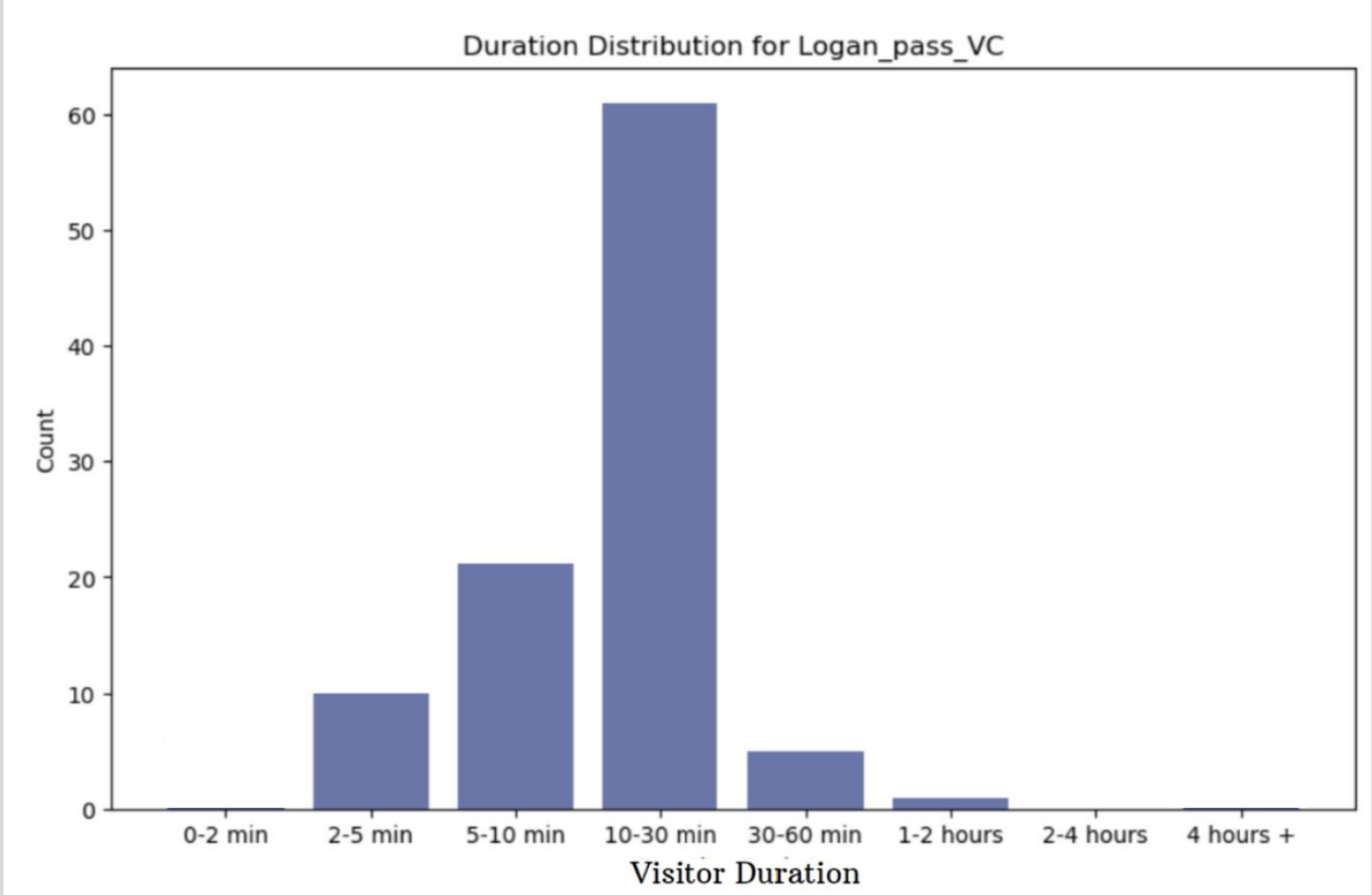
(N = 32)



Visitor Behavior at Logan Pass for Shuttle Arrivals

(N = 77)




Duration Distribution for Logan_pass_VC
Count
0
10
20
30
40
50
60
0-2 min
2-5 min
5-10 min
10-30 min
30-60 min
1-2 hours
2-4 hours
4 hours +
Visitor Duration

Staff Perception of Attraction to Logan Pass

Logan Pass Draws: Trails

“Once people are up there… they go out and do… some aspect of the trails, even if they weren’t planning to.” (Interview 3)



Data shows 75% of all Logan Pass visitors hike at least one trail there.

Logan Pass Draws: Animals



“Wildlife and hiking… are the two main things people are looking for (at Logan Pass)” –Interview 6

Logan Pass Draws: Infrastructure

“An awful lot of merchandise goes out of Logan Pass… they’re going straight into the store, they’re picking up their mementos of their trip and they’re heading on out”
-Interview 7

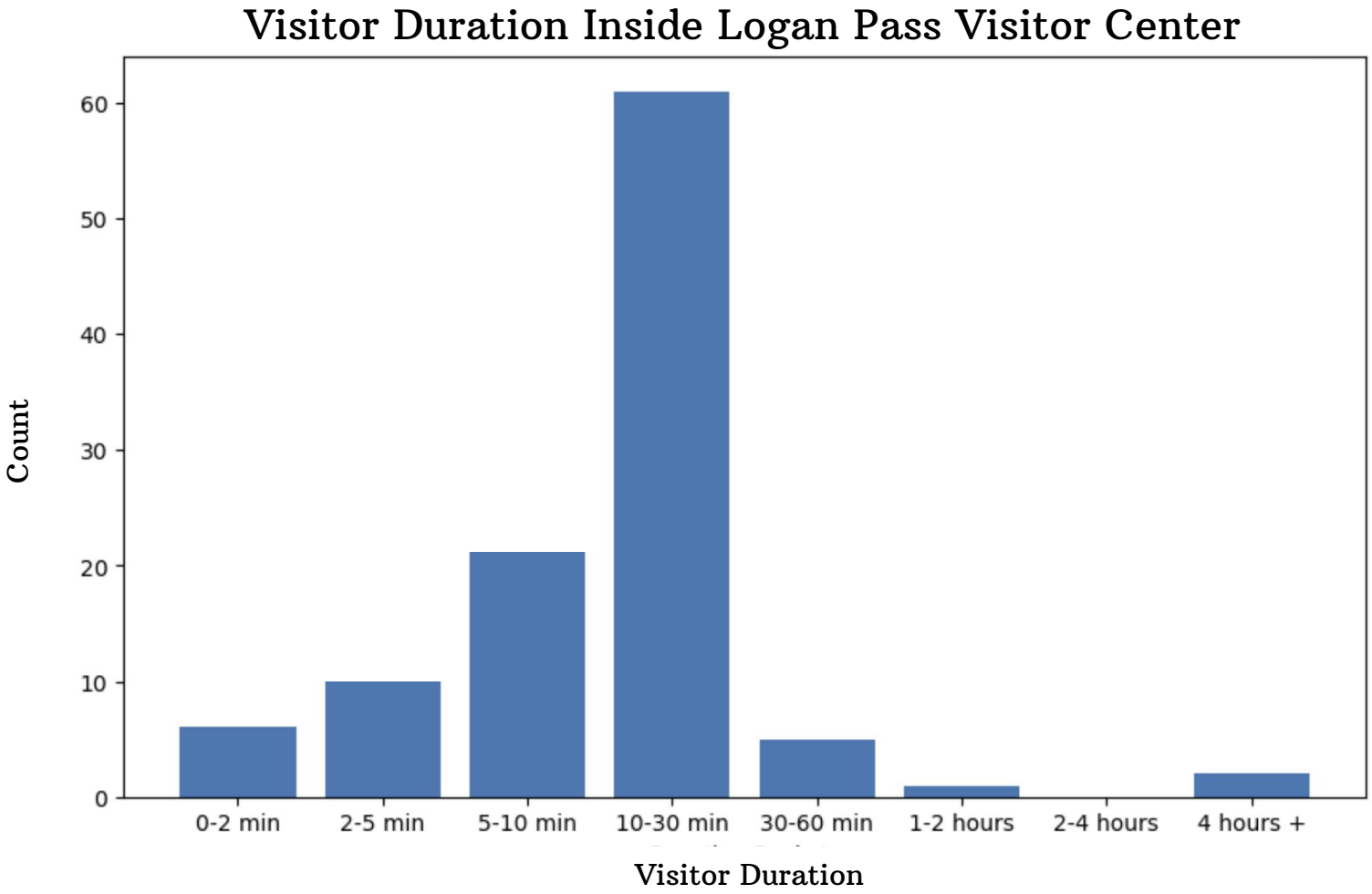


There are *“restroom lines… especially during peak summer. We do have a pretty large restroom up there, as far as regular flush toilets go…. It’s a good size, people just don’t like to wait”*
-Interview 2

Half of the visitors park rangers talk to are unaware of what Logan Pass has to offer.

“I would say at least half the people that I interact with at the visitor center or walking around the visitor center don’t know what’s there. They come up because that’s where they’ve been told or seen or read, that's where they should go, and are not aware of what’s even up there in terms of what is there to do, what are the trails?” (Interview 3)

Long duration at visitor centers implies that many people are seeking information about their trips.


Visitor Duration Inside Logan Pass Visitor Center
Count
60
50
40
30
20
10
0
0-2 min
2-5 min
5-10 min
10-30 min
30-60 min
1-2 hours
2-4 hours
4 hours +
Visitor Duration

Visitors are misinformed about the amenities or sites available at Logan Pass.

“There are the ones who used Instagram (and see a photo) and are like…, ‘that's so pretty.’ And they come up to us, and they’re like, ‘where’s this?’ And we’re like, ‘that’s 30 miles away and you need a backpacking permit.’” (Interview 2)

Many visitors, used to more developed national parks, “think there’s a restaurant up there…. They’re like, ‘where’s the cafeteria?’” (Interview 9)

“Other number one question we get is, ‘is there any coffee up here?’” (Interview 10)

Staff Perception of the Visitor Experience

Amenities in Logan Pass have long wait times and are congested during **peak** and **shoulder seasons** .

The shuttle service cannot keep up with the demand.

“People are sitting there in the sun … oftentimes waiting upwards of 45 minutes to get a shuttle [to Logan Pass]. A lot of [the shuttles] are already coming, driving past them full” (Interview 1).

The shoulder seasons are increasingly difficult for park rangers to maintain a high visitor experience.

Water is shut off, seasonal staff leaves, and amenities are closed in areas of the park.

“We are like at least a month into the shoulder season now… and it’s still very high visitation… It’s just becoming all main season until its not” (Interview 1).


Swiftcurrent Ridge
Lake
Boulder Ridge
Saint Mary Ridge
7409 ft
9400 ft
Adair Ridge
7968 ft
Boulder Creek
Dutch Ridge
Glacier Wall
Mary
Camas Ridge
Saint Mary River
Camas Creek
Howe Ridge
9277 ft
Lake McDonald
9381 ft
Snyder Ridge
Belton Hills
West Glacier
Harrison Lake
Nyack Creek
10122 ft
What's the big picture?

There is no immediate solution to seen in this data to improve visitor behavior—more work is necessary to better understand the complexities of the visitor experience.

"I think we know that there's… varied solutions, but they all take a lot of money and there would be a lot of work, and aren't perfect in figuring out… which we do."

"I think we need to figure out which problem we want to solve." (Interview 6)

Recommendations

- Continue working with park staff at Logan Pass to identify flaws in visitor experience.
- Conduct further research into methods of educating visitors.
- Conduct further research into the complex parking situation at Logan.

Expanding upon this study

- Creation of a transparent data cleaning protocol.
- Proper backing up procedure and labeling of GPS data.
- Times that accurately encapsulate visitor behavior.
- A GPS use tutorial so there is no separation between operators and different groups of study.
- Accurate Polygons and ARCGIS support files for proper data analysis.
- Clean, easy to understand ARCGIS scripture, for analysis and future use.
- Proper archiving of documents for future use.